INCH-POUND

MIL-C-24643/8C
14 March 1994
SUPERSEDING
MIL-C-24643/8B
3 October 1986

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 300 VOLTS, TYPE LSMMOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size 24, class O stranding, uncoated.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
* Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Standard identification code shall be applied by method 3.
Fourth - Five conductors cabled together with a lay not greater than 2 inches. A tie cord shall be laid straight as the central core. Fillers shall be used as necessary, to obtain a firm well-rounded assembly. Cabling sequence shall be consecutive.
Fifth - Binder of polyester tape or braid.
* Sixth - Cross-linked polyolefin jacket.

TABLE I. Details

Military part no. M24643/8	Type and size	Number of conductors	Insulated conductor minimum (inch)	Overall maximum (inch)
-01UN	LSMMOP-5	5	0.065	0.305

AMSC N/A FSC 6145

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EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	27.96
Voltage withstand - volts, root mean square, minimum..	1000
Insulation resistance - megohms/1000 feet, minimum....	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation.....	50
Jacket (When tested at 200°C).....	50
Tear - pounds per inch thickness, minimum (ASTM D 470).....	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.	250
Jacket (cord)	
Tensile strength - lb/in^2, minimum.	1300
Elongation - percent, minimum.....	160
Twisting and bending endurance -	
At 75°C, cycles, minimum.....	6000
At minus 30 ±2°C, cycles, minimum.....	2000
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged minimum.....	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged minimum.....	80

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	Jacket (cable)	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged minimum	60
	Hot oil immersion	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged minimum	50
	Shrinkage	No failure
*	Permanence of printing (jacket) - cycles, minimum	125
	Heat distortion - percent of unaged, maximum	30

Group D:

*	Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 500 feet nominal.

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